Digital Egypt Pioneers

وزارة الاتصالات
وتكنولوجيا المعلومات

Aegis AI

Version 1.0

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Introduction

The **Aegis AI** platform is designed to enhance public safety by providing automated, real-time detection of violence and distress in visual media. It addresses a wide spectrum of critical incidents, including Abuse, Arrest, Arson, Assault, Burglary, Explosion, Fighting, Road accidents, Robbery, Shooting, Shoplifting, Stealing, and Vandalism.

By leveraging cutting-edge deep learning architectures and computer vision techniques, Aegis AI transforms traditional surveillance from a passive recording tool into an active, intelligent response system. In an era where human-operated monitoring is limited by fatigue, attention span, and the sheer volume of data, our solution provides a persistent and objective analytical layer. The platform does not only identify these threats but also categorizes their severity and triggers immediate, context-aware actions to minimize harm and accelerate emergency response times.

Project Planning & Management

Project Proposal

Overview of the project

1. **Mission Statement**

Aegis AI is an intelligent visual monitoring platform designed to transform passive surveillance into an active safety infrastructure. By utilizing advanced Deep Learning techniques, the system provides autonomous, real-time detection and classification of violence and distress, ensuring a swift and organized response to critical incidents.

2. **The Core Solution**

The platform acts as a digital guardian that "understands" human actions within video streams. Instead of relying on manual observation, Aegis AI continuously scans visual data to identify 13 specific types of critical events, categorized into:

Violent Crimes: Assault, Fighting, Shooting, and Abuse.

Property Crimes: Robbery, Stealing, Burglary, Shoplifting, and Vandalism.

Public Emergencies: Road accidents, Explosions, Arson, and Arrests.

Datasets will used to train model were :

- Some Videos from YouTube
- XD-Violence
- CRCV | Center for Research in Computer Vision at the University of Central Florida
- RWF2000 - A Large Scale Video Database for Violence Detection

3. Key Functional Pillars

To achieve its goals, the system operates through three integrated layers:

Autonomous Detection: Utilizing Spatial-Temporal neural networks to recognize patterns of violence across consecutive video frames without human intervention.

Smart Categorization: A multiclass classification engine that identifies the specific nature of the threat to ensure the appropriate authority is notified.

Immediate Alerting: A centralized dashboard that provides instant visual and audio alerts to security personnel, minimizing the time between incident occurrence and first response.

4. Why Aegis AI?

Traditional security systems are often reactive—used primarily for reviewing footage after a crime has occurred. Aegis AI shifts the focus to Proactive Prevention. It solves the challenges of human fatigue and monitoring scalability, providing a reliable, 24/7 analytical layer that enhances the efficiency of security teams and increases public safety.

Project Objectives

1. **Automated Monitoring:** To develop a system capable of continuous, autonomous surveillance, reducing the burden on human operators and ensuring no critical events are missed due to fatigue or oversight.

2. **Precise Event Categorization:** To accurately classify different types of incidents to ensure that the appropriate response team is notified.

3. **Instant Alerting Framework:** To minimize the response time by providing immediate notifications to security personnel the moment a distress or violent event is detected.

4. **Centralized Reporting Dashboard:** To provide a user-friendly interface that displays live alerts, incident locations, and historical logs, making it easier for officials to review and document events.
5.

Project scope

1. In-Scope (Features to be Implemented)

This section outlines the core tasks and deliverables that the team is committed to delivering:

- **AI Model Development:** Building and training a Deep Learning model capable of processing the spatial and temporal dimensions of a video.
- **Action Classification:** Detects and classifies 13 types of accidents.
- **Real-time Processing:** The ability to analyze and process live streams or recorded video with the least possible latency.
- **Centralized Dashboard:** A dashboard for administrators that displays current cameras, real-time alerts, and incident logs.
- **Automated Alerting System:** A system to send instant notifications (visual and audible) to the control panel as soon as an emergency event is detected.
- **API Integration:** Building a software interface to connect the AI engine to the control panel and database.

2. Out-of-Scope (currently outside the project boundaries)

To ensure the team's focus, the project **will not** include the following:

- **Hardware Deployment:** The team is not responsible for installing physical cameras or laying networks at sites.
- **Biometric Identification:** The system will not do Face Recognition or Person Identification.
- **Audio Analysis:** Detection is based entirely on Visual Data only, and does not include analysis of sounds (such as gunshots or screams).
- **Government Integration:** Physical linking will not be done with police or ambulance databases; this will be replaced by an internal "simulated" alert system.
- **Mobile Application:** A web-based dashboard will be sufficed and does not currently include a mobile application.

3. System Constraints & Limitations

- **Lighting Conditions:** Detection quality may be affected in completely non-light conditions.

- **Camera Quality:** The minimum video processing is 480p quality to ensure the accuracy of Feature Extraction (it's one of the big cons).
- **Hardware Requirements:** The system requires a powerful GPU to ensure real-time operation.

Project Plan

Milestone

These are the points at which we measure progress, and they must be concluded on time:

- **M1: Data Completion:** Finish compilation and labeling of all videos (week 1).
- **M2: Model Prototype:** Train the first version of the model and reach acceptable accuracy (week 2).
- **M3: Full Integration:** Connecting the model to the backend and the Dashboard (week 4).
- **M4: Final Deployment:** Delivery of the final project and documentation (week 8).

So the project will take from three months to four months.

Dileveries

Tangible things that you will receive in your hand:

1. **Aegis AI Model:** The code for the AI model and trained weights.
2. **Security Dashboard:** A web app for displaying cameras and alerts.
3. **Action System API:** A software system for issuing notifications.
4. **Final Technical Report:** Comprehensive documentation of all stages of project construction.

Resource allocator

We are 6 members in the team and distributed on the stages of the project:

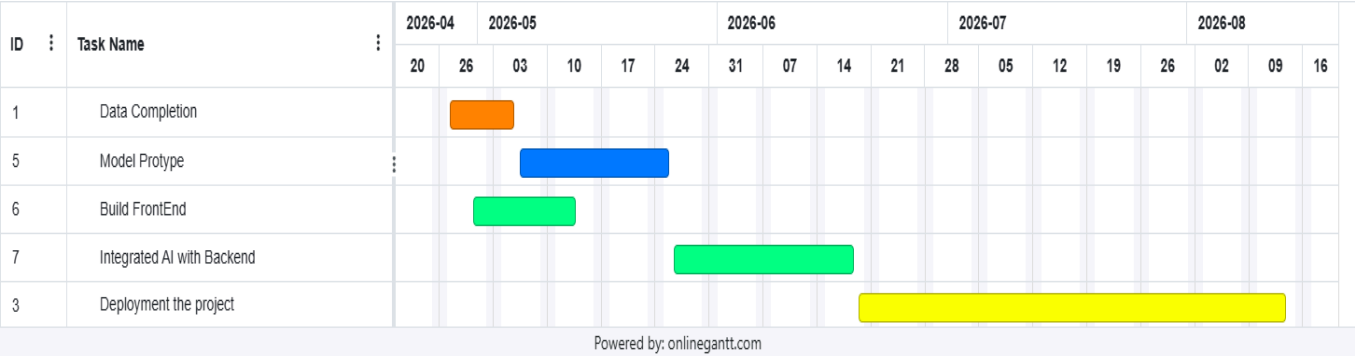
- **Project Leader (1 member - George):** Responsible for documentation, timeline, coordination among team members ,and sharing coding in the different stages.

- **Data Engineers (2 members - Asmaa & Nagham):** Video compilation, cleaning, annotation, and preprocessing.

- **AI/ML Engineers (2 members - Malak & Zainab):** Architectures, Training, Tuning, and Model Evaluation.

- **Full-Stack Developer (1 member - Mohamed):** Build the Dashboard (Frontend) and link it to the API (Backend) and Database.

Timeline



Task Assignment & Roles

To ensure maximum efficiency for our six members team, responsibilities are divided based on the **AI Pipeline** and **System Engineering** lifecycle. This structure prevents overlapping and ensures every component of **Aegis AI** is handled efficiently.

1. Individual Roles & Responsibilities

Role 1: Project Lead & System Architect

- **Owner:** ***George***
- **Key Responsibilities:**
  - Defining the project roadmap, milestones, and technical requirements.
  - Determining the roles of each member and his/her responsibilities.
  - Overseeing **System Integration** to ensure the AI model, Backend, and Frontend communicate seamlessly.

    - Managing technical documentation and the final project report.

Role 2 & 3: AI & Deep Learning Engineers

- **Owners:** ***Malak & Zainab***
- **Key Responsibilities:**
    - Researching and implementing the model architecture.
    - Managing **Model Training** and hyperparameter tuning on GPU environments.
    - Performing **Model Evaluation** using metrics such as Confusion Matrix, Precision, and Recall.
    - Optimizing the model for **Real-time Inference** to minimize processing latency.

Role 4 & 5: Computer Vision & Data Engineers

- **Owners:** ***Asmaa & Nagham***
- **Key Responsibilities:**
    - Sourcing, cleaning, and organizing the **Video Datasets**.
    - Building the **Preprocessing Pipeline** (Frame extraction, Normalization, and Resizing).
    - Applying **Data Augmentation** techniques to balance classes.
    - Implementing **Optical Flow** or temporal feature extraction to enhance motion recognition.

Role 6: Full-Stack Systems Developer

- **Owner:** *Mohamed*
- **Key Responsibilities:**
    - **Backend:** Developing the server (FastAPI/Flask) to handle video streams and AI predictions.
    - **Database:** Designing the schema to log incidents, timestamps, and alert history.
    - **Frontend:** Building the **Aegis Dashboard** for real-time monitoring and visual alerts.
    - Implementing the **Action System** (automated notifications and UI triggers).

Risk Assessment & Mitigation Plan

Here we determine the technical "piles" that can be met by the 6 members, and how to solve them before they happen.

Risk	Type	Impact	Mitigation Plan

Data Scarcity	Technical	high	Rely on **Transfer Learning** from ready-made models, and use **Data Augmentation** to .generate additional images
High Latency	Technical	high	Use **model quantization** or **pruning** techniques to reduce model size, reducing the number of frames processed per second .((FPS
Hardware Limitations	Resources	Medium	Training with cloud platforms such as **Google Colab** or **Kaggle**, and using images with .(smaller sizes (such as 128×128
Data Collection	Resources	Medium	Download from YouTube and other websites Videos that make larger dataset.
False Positives	Technical	Medium	Training the model on "negative" data is a lot like videos of people playing a sport or joking .to learn the difference
Integration Failure	Administrative	high	Define the **API Contract** (the form of data that smells) from the first week between the .AI team and the full-stack team

KPIs (Key Performance Indicators)

These are the numbers that will give us where we are ?

A. Technical Metrics:

- **Detection Accuracy:** Achieving at least **85%** accuracy in classifying accidents on the Test Set.
- **Inference Time:** The video is processed and the resolution appears in less **than 500ms** (half a second) to ensure real-time.
- **Recall Rate:** We are very interested in Recall (MivoTech means a real incident) and it targets **90%** of serious incidents (such as shooting).

B. Operational Metrics:

- **Alert Lead Time:** The time from the moment of the incident to the arrival of the alert to security on the dashboard is less **than 2 seconds**.
- **System Uptime:** Server and dashboard stability to **work 99%** without hassle while processing streams.

C. Data Metrics:

- **Class Balance:** Ensure that there are at least **500+ videos** for each of the 13 types to ensure that the model is not biased.

Gentle Note :

To ensure project success, all roles and deadlines are treated as living documents. They are subject to change based on the project's evolving needs and real-time development insights.

Thanks DEPI